

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			Any 4 (x1) from: Similarities: A. Have phospholipid bilayer/ hydrophobic tails & hydrophilic heads (1) B. Correct location of heads and tails: (hydrophobic tails) inside & (hydrophilic heads) outside (1) C. Have protein (1) Differences: D. {Proteins not embedded in the lipid bilayer / proteins coat outer surface / continuous layer of protein / no intrinsic proteins} in Davson Danielli model (1) E. Ref to absence of {glycocalyx/ glycoproteins/ glycolipids/ cholesterol} in Davson Danielli model (1) Accept reverse for Fluid Mosaic		4		4		
	(b)	(i)		A. As the number of carbon atoms in the chain increases, the (transition) temperature increases (1) Accept increased chain length B. Because the longer the chain length the greater the intermolecular forces (1) Accept increased number of C atoms C. Therefore more energy needed to break the {bonds / overcome the intermolecular forces}(1) Accept reverse argument			3	3		

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		(ii)		Unsaturated contain {double bonds between carbons/ C=C}(1) Any 3 (x1) from: • Produces a kink / bend, in the side chain (1) • Less packing is possible/ increased distance between the chains (1) • Lowers intermolecular force /(phase) transition temperature (1) • less energy needed {to break the bonds/ overcome the intermolecular forces} (1)	1		3	4		
		(iii)		Between the {fatty acids / hydrophobic tails} Accept labelled diagram		1		1		
		(iv)		Raises level of LDLs/ causes LDLs to be made (1) Increase incidence of atheroma in arteries/ detailed description of atheroma (1)	2			2		
				Question 5 total	3	5	6	14	0	0